

Inter-thread document overlap (%) — Jaccard similarity over thread doc-sets

MHQA — Naive parallel

Qwen3-1.7B	16	63	20	74	58
Qwen3-4B	64	78	57	82	77
Qwen3-8B	76	63	71	77	62
Gemma3-4B	—	42	—	57	43
Gemma3-12B	14	44	29	61	49
	HpQA	MuSi	2Wiki	Bambo	FRAMES

MHQA — DIFFUSE

Qwen3-1.7B	35	32	32	36	35
Qwen3-4B	33	27	28	31	29
Qwen3-8B	26	24	24	29	26
Gemma3-4B	5	9	4	13	13
Gemma3-12B	8	14	6	20	21
	HpQA	MuSi	2Wiki	Bambo	FRAMES

Reasoning — Naive parallel

Qwen3-4B	40	35	51
Qwen3-8B	42	35	52
Gemma3-12B	31	21	31
Gpt-oss-20B	—	—	2
	GAIA	HLE	WebW

Reasoning — DIFFUSE

Qwen3-4B	22	17	24
Qwen3-8B	18	15	20
Gemma3-12B	14	10	16
Gpt-oss-20B	6	2	8
	GAIA	HLE	WebW

Cells with fewer than 100 valid trajectories are masked (—); they reflect runs where naive parallel produced few successful searches.